

# Python Interview Questions

1. What is a variable in Python?
2. Difference between dynamically typed and statically typed language?
3. How do you check the data type of a variable?
4. Difference between int, float, complex, and bool?
5. Is Python case-sensitive?
6. What is the difference between mutable and immutable data types?
7. Which Python data types are immutable?
8. What happens internally when you write: `a = b = 10`
9. What is typecasting?
10. Difference between implicit and explicit type conversion?
11. Output: `print(int(10.8))`
12. Can we convert string to integer always?
13. Difference between `str(10)` and `repr(10)`?
14. Why are strings immutable in Python?
15. Difference between single quotes, double quotes, and triple quotes?
16. Output: `a = 'Python' ; print(a[1:5])`
17. Difference between `find()` and `index()`?
18. Difference between `split()` and `join()`?
19. Output: `print('hello' * 3)`
20. How do you reverse a string without using built-in functions?
21. Difference between `==` and `is` for strings?
22. Difference between List and Tuple?
23. Why is Set unordered?
24. Difference between Dictionary keys and values?
25. Difference between `append()`, `extend()`, and `insert()` in List?

## Bonus Output Questions

Q1

```
a = [1,2,3]  
b = a  
b.append(4)
```

```
print(a)
```

Q2

```
print(bool([]))  
print(bool([1]))
```

Q3

```
a = (1)  
b = (1,)
```

```
print(type(a))  
print(type(b))
```

Q4

```
print(10 / 2)  
print(10 // 2)
```

Q5

```
print(2 ** 3)
```